Enabling Onboard Navigation for UAVs in GPS Denied Environments

MIP400A : B.Tech Project - MidTerm Evaluation

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Objectives

Goal: UAV self-localization when GPS/GNSS is unavailable, jammed, or spoofed, using only onboard sensors.

- **This semester:** working end-to-end pipeline, in simulation.
- **Next semester (through Apr 2027):** fast, power-efficient (FPGA) onboard deployment.

Core approach: Scene-Aided Navigation

Replace GPS with passive measurements the aircraft can already sense: terrain, gravity, magnetic field, and visual scene, matched against a reference map.

Motivation

Weak signal

GNSS is easily jammed, spoofed, or simply unavailable.

Mechanical view

The IMU already senses acceleration, rotation, and gravity; a camera could replace the satellite link entirely.

Must keep flying

Survey, disaster-response, and defense UAVs need to operate even when GPS fails.

Precedent

TERCOM and DSMAC proved passive, GPS-free terrain and scene matching decades ago.

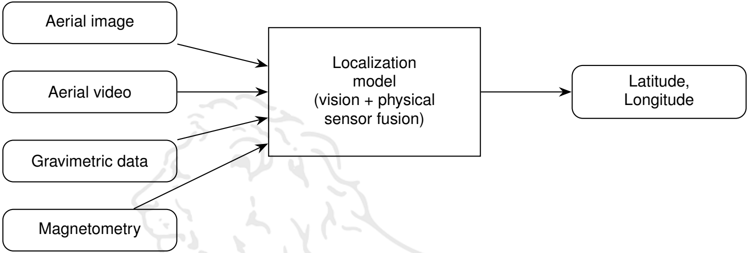
Background of Similar Work

Passive GPS-denied navigation methods, screened for feasibility on a small UAV:

Method	Physical quantity measured	Passive?	Assessment
Terrain matching (TERCOM)	Ground elevation	Yes	Proven, strong candidate[4,6]
Visual landmark matching	Camera imagery	Yes	Strong if paired with AI matching[1-3,5,7,12,13]
Transformer-based matching	Camera imagery	Yes	SOTA in this niche; motivates our redesign[14,15]
Gravity matching	Gravity anomalies	Yes	Promising, needs sensitive sensor[8,9]
Gravity-gradient nav.	$\partial g/\partial x, y, z$	Yes	Very promising, higher precision[8,9]
Magnetic anomaly nav.	Magnetic field	Yes	Useful in mountainous terrain[10,11]
Quantum inertial sensing	Acceleration / rotation	Yes	No map needed; pure mechanics
Wi-Fi / cellular / RF	Radio signal strength	No	Rejected: unavailable at altitude

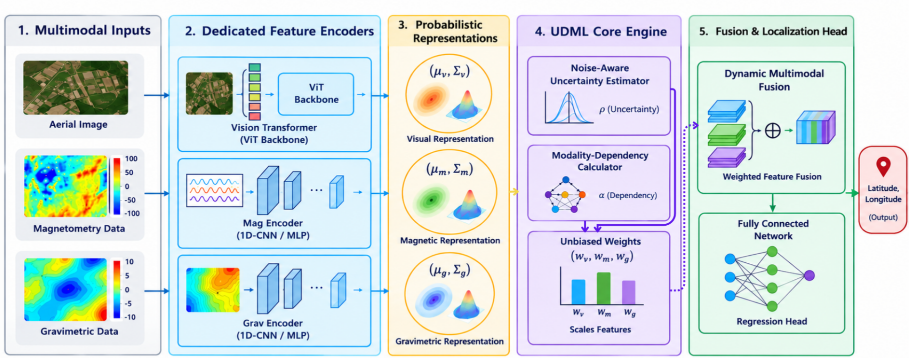
Reviewed: VGG16[1], CLIP[2,3,7], SIFT/ORB[5], and DEM terrain-weighting[4,6], achieving sub-3–7 m RMSE[2,4].

Proposed Work



- **Baseline validated:** replicated the reference implementation[1] in simulation; accuracy matches the paper's reported numbers.
- **Now:** redesigning the backbone as a **transformer** (next slide), trading compute and latency for higher accuracy, then adding multimodal sensor fusion.

Proposed Model Architecture



- **Transformer vision backbone** (ViT) replaces VGG16[1]; magnetometry and gravimetry get their own 1D-CNN/MLP encoders.
- Each modality is encoded as a **probabilistic representation** (μ, Σ), so sensor noise is carried forward explicitly.
- The **UDML core** estimates per-modality uncertainty ρ and dependency α, producing unbiased fusion weights (w_v, w_m, w_g).
- Weighted features are fused and regressed to **latitude, longitude** by a fully connected head.

Timeline of the Work

Phase 1: Now to Nov 2026

Transformer backbone + multimodal fusion (IMU, magnetometer, gravimeter)

end-to-end pipeline, in simulation

Phase 2: Jan to Apr 2027

Optimize for speed & power on FPGA

physical UAV test

Phase 2 extends our lab project from last semester, which showed FPGAs are highly power-efficient for this class of computation.

References (1/2)

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[2] *ngps_flight*: real-time CLIP-style UAV–satellite matching, ∼3 m RMSE on Jetson Orin. https://github.com/snktshrma/ngps_flight

[3] *CLIP-UAV-localization*: CLIP-based network matching UAV imagery, also recovers heading. https://github.com/codebra721/CLIP-UAV-localization

[4] *GNSS-Denied-UAV-Geolocalization*: terrain-weighted optimization using satellite maps and DEM data, sub-7 m. https://github.com/YFS90/GNSS-Denied-UAV-Geolocalization

[5] *VisionUAV-Navigation*: classical SIFT/ORB hybrid feature detection, no deep learning. https://github.com/sidharthmohannair/VisionUAV-Navigation

[6] GNSS-denied UAV geolocalization via terrain-weighted constraint optimization, sub-7 m error, *ScienceDirect*. https://www.sciencedirect.com/science/article/pii/S1569843224006332

[7] CLIP-based UAV–satellite image matching, also derives heading direction, *ScienceDirect*. https://www.sciencedirect.com/science/article/pii/S2590123025041787

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[14] X. Zhu et al., *TransGeo: Transformer Is All You Need for Cross-view Image Geo-localization*, CVPR 2022. https://arxiv.org/abs/2204.00097

[15] *FSRA*: A Transformer-Based Feature Segmentation and Region Alignment Method for UAV-View Geo-Localization. https://arxiv.org/pdf/2201.09206